

Yiquan Wang

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Memberships: Biophysical Society of China • Chinese Chemical Society • IEEE Biometrics Council

Education

Xinjiang University (National Base for Mathematical Research and Teaching Talents)

Bachelor of Science in Mathematics and Applied Mathematics

Urumqi, Xinjiang

2023.09 – 2027.06

Tsinghua University & X-Institute

Tsien Excellence in Engineering Program (Joint Program, X Scholar)

Shenzhen, Guangdong

2024.06 – 2027.06

Research Experience

Beijing Frontier Research Center for Biological Structure, Tsinghua University

Research Intern | Supervised by Prof. Yafei Yuan & Prof. Tong Wang

Beijing, China

2025.07 – Present

- Engaged in structure-based protein and polypeptide design research leveraging generative Artificial Intelligence.
- Investigated protein conformational changes and functional mechanisms through AI-driven molecular dynamics (MD) simulations.
- Independently developed high-throughput data infrastructures for the structural biology community: the [Peptide Design Competition Website](#) and the [Polypeptide Structure Database](#).

Shenzhen Bay Laboratory

Visiting Student | Institute of Neurological and Psychiatric Disorders (Wen Yuan's Group)

Shenzhen, Guangdong

2025.07 – 2025.09

- Engaged in interdisciplinary research across computational biology, neuroscience, and biochemistry, integrating high-throughput single-cell multi-omics and CRISPR screening to investigate the pathogenic mechanisms and therapeutic targets of neurological disorders (e.g., Alzheimer's and autism).

AI-Driven Protein Function Prediction & Generative Design (Tsinghua University ESRT)

2024.08 – 2025.10

Project Lead | Supervised by Prof. Kai Wei

- Proposed protein sonification (1D to 2D spectrograms); built a multimodal fusion model achieving 90.44% accuracy on CARE enzyme benchmark, matching standard transformers (ESM-2, ProtBERT) in data efficiency.
- Guided a Diffusion Model using the novel Two-Dimensional encoding to generate Green Fluorescent Protein variants, validated via biophysics.

Selected Publications

AI-Driven Protein Modeling, Design & Structural Bioinformatics

- Wang, Y., Cai, M., Dong, Y., Ma, Y., & Wei, K. (2025). [From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction](#). *Journal of Chemical Information and Modeling*, 65(23), 12723-12736. (JCR Q1, CAS Q2 Top).
- Wang, Y., Cai, M., Ma, Y., & Wei, K. (2026). [Sequence-context-aware decoding enables robust reconstruction of protein dynamics from crystallographic B-factors](#). Accepted to *Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026*, under review at *Science China Life Sciences*.
- Wang, X.[†], Wang, Y.[†], Huang, T.-Y., Dong, Y., Deng, J., Xu, L., Li, X., & He, R. (2025). [Octopus Inspired Optimization \(OIO\): A Hierarchical Framework for Navigating Protein Fitness Landscapes](#). *2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 6897-6904. (CCF-B, Co-first author).
- Wang, Y., Ma, Y., Chang, Y., Yan, J., Zhang, J., Cai, M., & Wei, K. (2025). [Diffusion Models at the Drug Discovery Frontier: A Review on Generating Small Molecules Versus Therapeutic Peptides](#). *Biology*, 14(12), 1665. (JCR Q1, CAS Q2).

First-Principles of Protein Geometry via Mathematical Physics

- Wang, Y. (2026). [Spectral entropy of the discrete Hasimoto effective potential exposes sub-residue geometric transitions in protein secondary structure](#). *arXiv preprint arXiv:2602.21787*.
- Wang, Y. (2026). [Piecewise integrability of the discrete Hasimoto map for analytic prediction and design of helical peptides](#). *arXiv preprint arXiv:2602.16255*.
- Wang, Y. (2026). [Structural barriers of the discrete Hasimoto map applied to protein backbone geometry](#). *arXiv preprint arXiv:2602.13160*.

Selected Awards & Academic Engagement

- Selected Awards:** Synthetic Biology Challenges Silver Award × 2 (2025); Contemporary Undergraduate Mathematical Contest in Modeling National Second Prize (2025); Mathematical Contest in Modeling Honorable Mention (2025).
- Academic Engagement:** Tsinghua-Peking University Center for Life Sciences Summer Camp (2025); Brown University Department of Physics Artificial Intelligence Winter School (2025).